

LAPORAN PRAKTIKUM ASD



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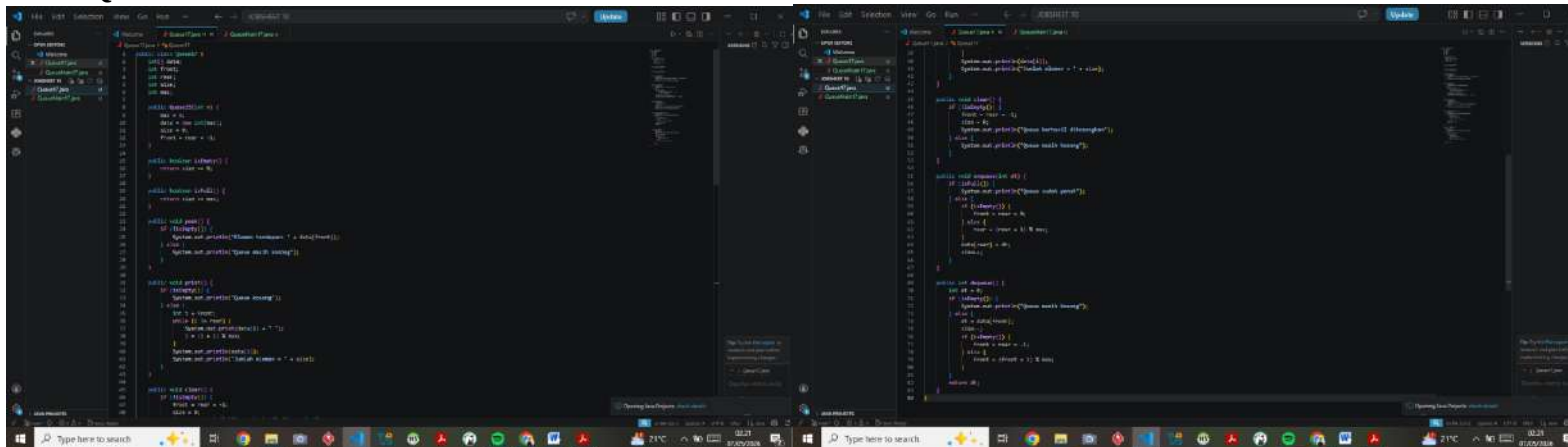
Sistem Informasi Bisnis / 1E

JOBSHEET 10 QUEUE

Tujuan Praktikum

1. Mengenal struktur data Queue.
2. Membuat dan mendeklarasikan struktur data Queue.
3. Menerapkan algoritma Queue menggunakan array.

Praktikum 1 – Operasi Dasar Queue Class Queue



```
class Queue {
    private int[] array;
    private int front;
    private int rear;
    private int size;

    public Queue(int capacity) {
        array = new int[capacity];
        front = -1;
        rear = -1;
        size = capacity;
    }

    public boolean isEmpty() {
        return front == -1;
    }

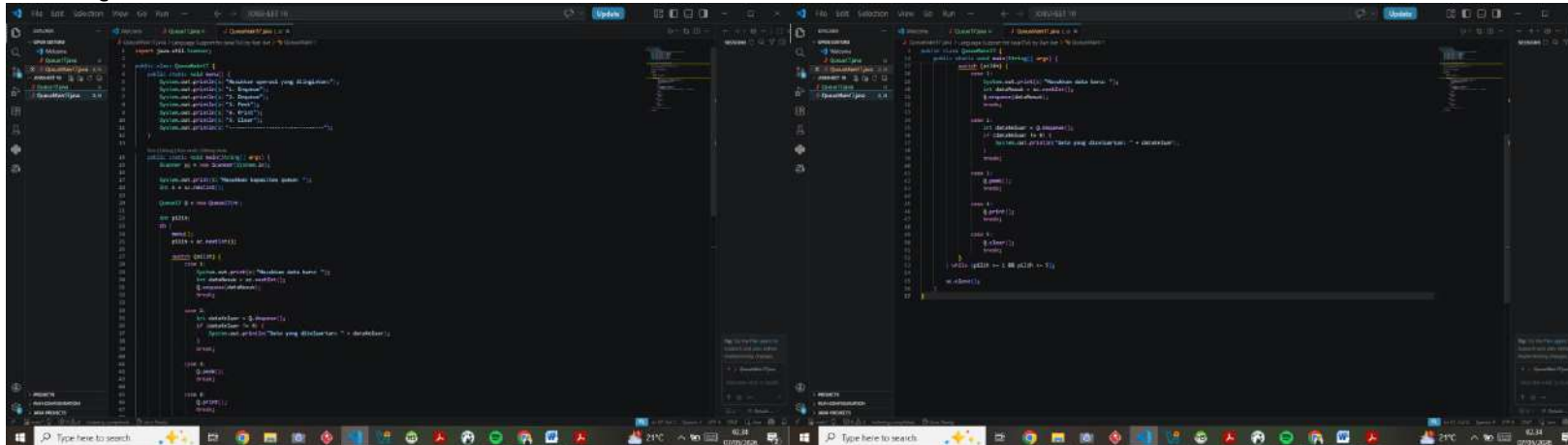
    public boolean isFull() {
        return rear == size - 1;
    }

    public void enqueue(int data) {
        if (isFull()) {
            System.out.println("Queue is full!");
            return;
        }
        if (isEmpty()) {
            front = 0;
        }
        rear++;
        array[rear] = data;
        System.out.println("Data " + data + " added to queue.");
    }

    public int dequeue() {
        if (isEmpty()) {
            System.out.println("Queue is empty!");
            return -1;
        }
        int data = array[front];
        front++;
        System.out.println("Data " + data + " removed from queue.");
        return data;
    }
}

class QueueMain {
    public static void main(String[] args) {
        Queue queue = new Queue(5);
        queue.enqueue(10);
        queue.enqueue(20);
        queue.enqueue(30);
        queue.enqueue(40);
        queue.enqueue(50);
        queue.dequeue();
        queue.dequeue();
        queue.dequeue();
        queue.dequeue();
        queue.dequeue();
    }
}
```

Class QueueMain



```
class QueueMain {
    public static void main(String[] args) {
        Queue queue = new Queue(5);
        queue.enqueue(10);
        queue.enqueue(20);
        queue.enqueue(30);
        queue.enqueue(40);
        queue.enqueue(50);
        queue.dequeue();
        queue.dequeue();
        queue.dequeue();
        queue.dequeue();
        queue.dequeue();
    }
}
```

The screenshot displays a Windows desktop with a Visual Studio Code editor open. The editor shows a C++ program for a queue implementation. The program includes a header file `QUEUE.H` and a source file `QUEUE.cpp`. The `QUEUE.H` file defines a `QUEUE` struct and a `QUEUE` class. The `QUEUE` class has a static array `arr` of size `10` and a static variable `front` of type `int`. The `QUEUE` class has three methods: `enqueue`, `dequeue`, and `display`. The `enqueue` method adds an element to the queue, the `dequeue` method removes an element from the queue, and the `display` method displays the elements of the queue. The `main` function calls these methods for three test cases.

```

1 #include <iostream>
2 #include "QUEUE.H"
3 using namespace std;
4
5 int main()
6 {
7     QUEUE q;
8     q.enqueue(1);
9     q.enqueue(2);
10    q.enqueue(3);
11    q.enqueue(4);
12    q.enqueue(5);
13    q.enqueue(6);
14    q.enqueue(7);
15    q.enqueue(8);
16    q.enqueue(9);
17    q.enqueue(10);
18    q.dequeue();
19    q.dequeue();
20    q.dequeue();
21    q.dequeue();
22    q.dequeue();
23    q.dequeue();
24    q.dequeue();
25    q.dequeue();
26    q.dequeue();
27    q.dequeue();
28    q.dequeue();
29    q.display();
30    return 0;
31 }

```

The output window shows the execution results for three test cases. The first test case shows the queue after enqueuing 10 elements and dequeuing 10 elements. The second test case shows the queue after enqueuing 10 elements and dequeuing 5 elements. The third test case shows the queue after enqueuing 10 elements and dequeuing 5 elements.

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```

Program ini mengimplementasikan struktur data Queue (antrian) menggunakan array.

- Enqueue() → menambah data ke belakang queue.
- Dequeue() → menghapus data dari depan queue.
- peek() → melihat elemen terdepan.
- print() → menampilkan seluruh isi queue.
- clear() → menghapus semua isi queue.
- IsEmpty() → cek queue kosong.
- IsFull() → cek queue penuh.

Jawaban Pertanyaan Praktikum 1

Karena queue awalnya kosong.

- `front = -1` artinya belum ada elemen depan.
- `rear = -1` artinya belum ada elemen belakang.

- `size = 0` karena jumlah elemen masih nol.

2. Jelaskan potongan kode Enqueue

```
if (rear == max - 1) {  
    rear = 0;  
} else {  
    rear++;  
}
```

Kode ini digunakan untuk **circular queue**.

Jika rear sudah di indeks terakhir array, maka kembali ke indeks 0 agar array bisa dipakai ulang.

3. Jelaskan potongan kode Dequeue

```
if (front == max - 1) {  
    front = 0;  
} else {  
    front++;  
}
```

Digunakan agar front berpindah maju setelah data dihapus.

Jika front sudah di indeks terakhir, maka kembali ke indeks 0.

4. Mengapa print mulai dari front?

Karena queue menampilkan data dari elemen pertama yang aktif.

Kalau mulai dari 0, bisa ikut menampilkan data lama yang sudah tidak termasuk queue.

5. Jelaskan kode berikut

```
i = (i + 1) % max;
```

Digunakan agar indeks bergerak circular.

Contoh:

`max = 5`

`i = 4`

Maka:

`(4+1)%5 = 0`

Jadi kembali ke awal array.

6. Potongan kode queue overflow

```
if (IsFull () ) {  
    System.out.println ("Queue sudah penuh") ;  
}
```

Overflow terjadi saat queue penuh tetapi masih ditambah data.

7. Modifikasi agar program berhenti saat overflow/underflow

Tambahkan:

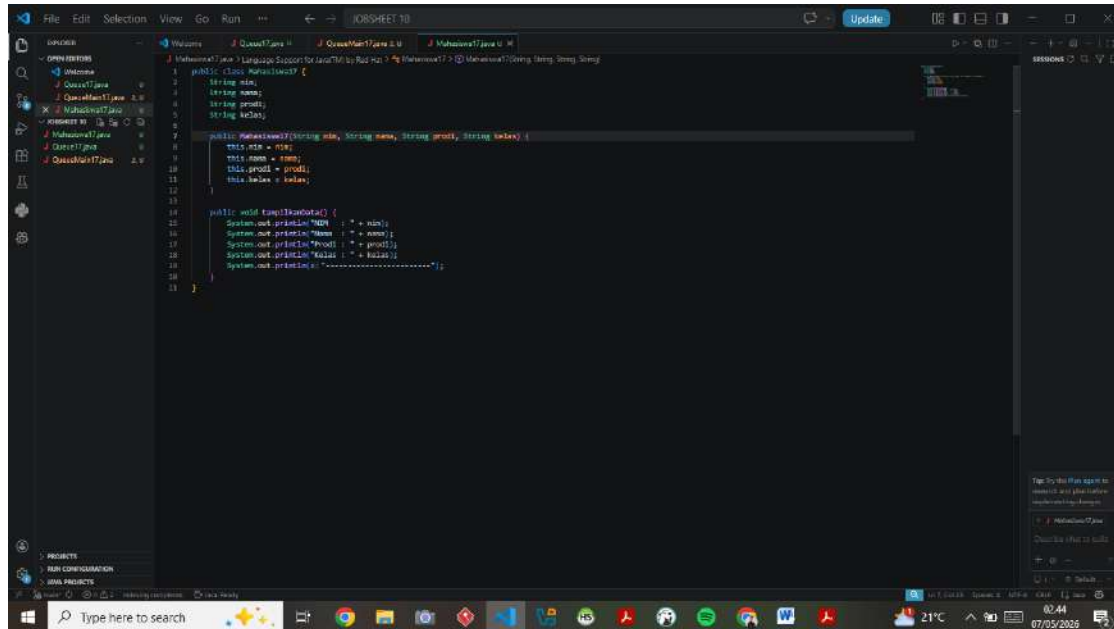
System.exit (0) ;

Pada kondisi:

queue penuh

queue kosong saat dequeue

Praktikum 2 – Antrian Layanan Akademik Class Mahasiswa



Class AntrianLayanan

The image shows two side-by-side IDE windows displaying the implementation of the `AntrianLayanan` class. The left window shows the `main` method, which creates an instance of `AntrianLayanan` and calls its `mainMenu` method. The right window shows the `mainMenu` method, which uses a `switch` statement to handle different menu options. The code is written in Java and includes comments in Indonesian.

```
import java.util.Scanner;

public class AntrianLayanan {
    private static Scanner scanner = new Scanner(System.in);

    public void mainMenu() {
        System.out.println("=====");
        System.out.println("1. Mendaftar");
        System.out.println("2. Mengambil Nomor");
        System.out.println("3. Menunggu");
        System.out.println("4. Keluar");
        System.out.println("=====");
        System.out.println("Pilih menu: ");
        int pilihan = scanner.nextInt();

        switch (pilihan) {
            case 1:
                pendaftaran();
                break;
            case 2:
                ambilNomor();
                break;
            case 3:
                tunggu();
                break;
            case 4:
                keluar();
                break;
            default:
                System.out.println("Pilihan tidak valid");
        }
    }

    private void pendaftaran() {
        System.out.println("Mendaftar");
        System.out.println("Nama: ");
        String nama = scanner.nextLine();
        System.out.println("No. Pendaftaran: ");
        int noPendaftaran = scanner.nextInt();
        System.out.println("Pendaftaran berhasil");
    }

    private void ambilNomor() {
        System.out.println("Mengambil Nomor");
        System.out.println("No. Pendaftaran: ");
        int noPendaftaran = scanner.nextInt();
        System.out.println("Nomor yang diambil: " + noPendaftaran);
    }

    private void tunggu() {
        System.out.println("Menunggu");
        System.out.println("No. Pendaftaran: ");
        int noPendaftaran = scanner.nextInt();
        System.out.println("Menunggu...");
    }

    private void keluar() {
        System.out.println("Keluar");
        System.out.println("No. Pendaftaran: ");
        int noPendaftaran = scanner.nextInt();
        System.out.println("Keluar");
    }
}
```

Class LayananAkademikSIKAD

The image shows two side-by-side IDE windows displaying the implementation of the `LayananAkademikSIKAD` class in Java. The left window shows the `main` method, which creates an instance of `LayananAkademikSIKAD` and calls its `mainMenu` method. The right window shows the `mainMenu` method, which uses a `switch` statement to handle different menu options. The code is written in Java and includes comments in Indonesian.

```
import java.util.Scanner;

public class LayananAkademikSIKAD {
    private static Scanner scanner = new Scanner(System.in);

    public void mainMenu() {
        System.out.println("=====");
        System.out.println("1. Mendaftar");
        System.out.println("2. Mengambil Nomor");
        System.out.println("3. Menunggu");
        System.out.println("4. Keluar");
        System.out.println("=====");
        System.out.println("Pilih menu: ");
        int pilihan = scanner.nextInt();

        switch (pilihan) {
            case 1:
                pendaftaran();
                break;
            case 2:
                ambilNomor();
                break;
            case 3:
                tunggu();
                break;
            case 4:
                keluar();
                break;
            default:
                System.out.println("Pilihan tidak valid");
        }
    }

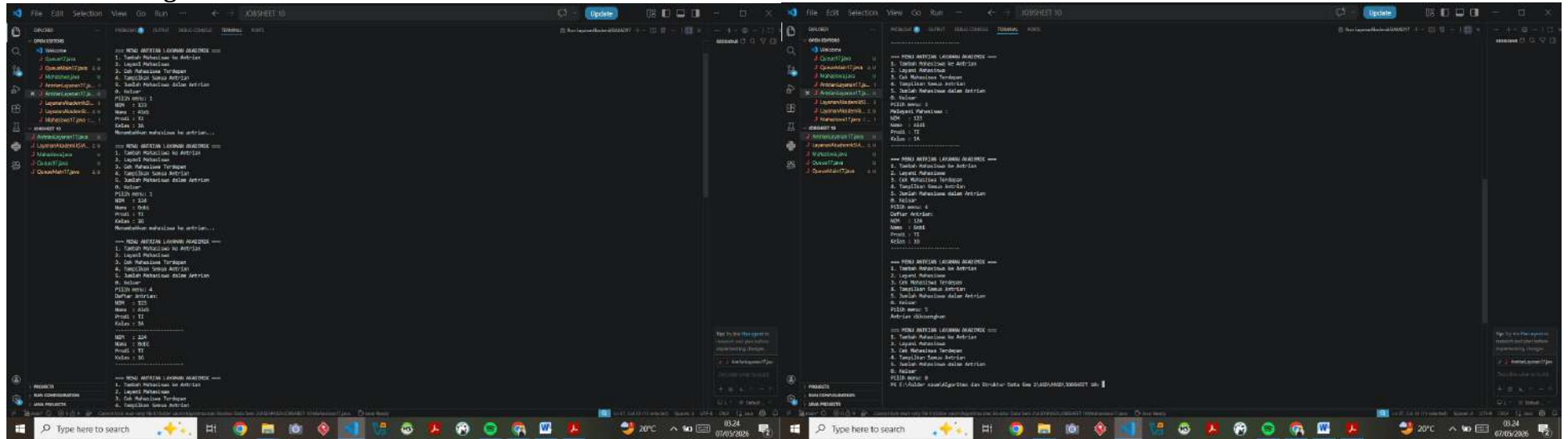
    private void pendaftaran() {
        System.out.println("Mendaftar");
        System.out.println("Nama: ");
        String nama = scanner.nextLine();
        System.out.println("No. Pendaftaran: ");
        int noPendaftaran = scanner.nextInt();
        System.out.println("Pendaftaran berhasil");
    }

    private void ambilNomor() {
        System.out.println("Mengambil Nomor");
        System.out.println("No. Pendaftaran: ");
        int noPendaftaran = scanner.nextInt();
        System.out.println("Nomor yang diambil: " + noPendaftaran);
    }

    private void tunggu() {
        System.out.println("Menunggu");
        System.out.println("No. Pendaftaran: ");
        int noPendaftaran = scanner.nextInt();
        System.out.println("Menunggu...");
    }

    private void keluar() {
        System.out.println("Keluar");
        System.out.println("No. Pendaftaran: ");
        int noPendaftaran = scanner.nextInt();
        System.out.println("Keluar");
    }
}
```

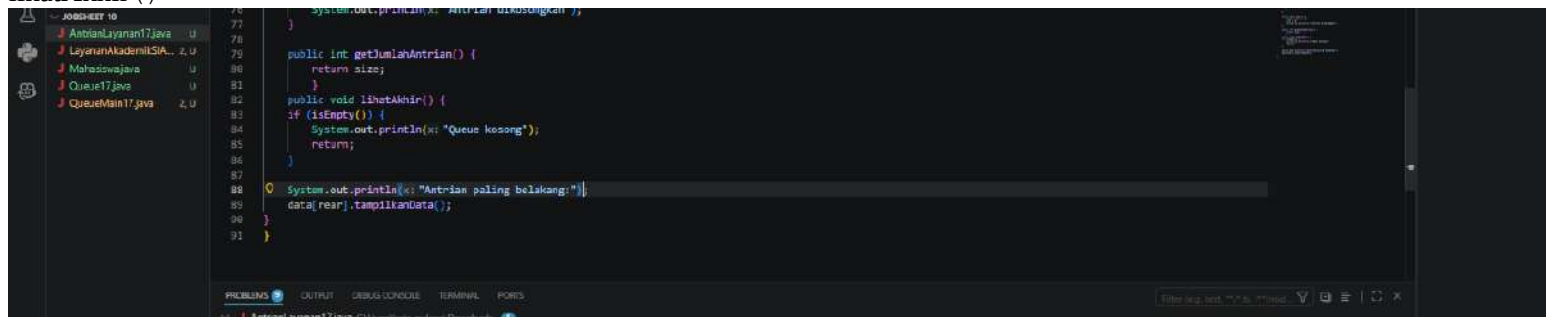
Hasil Run Program



4.3 Jawaban Pertanyaan Praktikum 2

Tambahkan method LihatAkhir

Sudah ditambahkan:
lihatAkhir ()



Menu:

6. Cek antrian belakang



```
1 import java.util.Scanner;
2
3 public class LayananAkademikSIKAD17 {
4     public static void Menu() {
5         System.out.println("\n== MENU ANTRIAN LAYANAN AKADEMIK ==");
6         System.out.println("1. Tambah Mahasiswa ke Antrian");
7         System.out.println("2. Layani Mahasiswa");
8         System.out.println("3. Cek Mahasiswa Terdepan");
9         System.out.println("4. Tampilkan Semua Antrian");
10        System.out.println("5. Jumlah Mahasiswa dalam Antrian");
11        System.out.println("6. Cek Antrian Paling Belakang");
12        System.out.println("0. Keluar");
13        System.out.print("Pilih menu: ");
14    }
15
16    Run (Debug) Run main (Debug main)
17    public static void main(String[] args) {
18        Scanner sc = new Scanner(System.in);
19
20        AntrianLayanan17 antrian = new AntrianLayanan17(10); // kapasitas antrian default 10
21
22        int pilih;
23        do {
24            menu();
25        } while (pilih != 0);
26    }
27 }
```

Switch:

case 6:

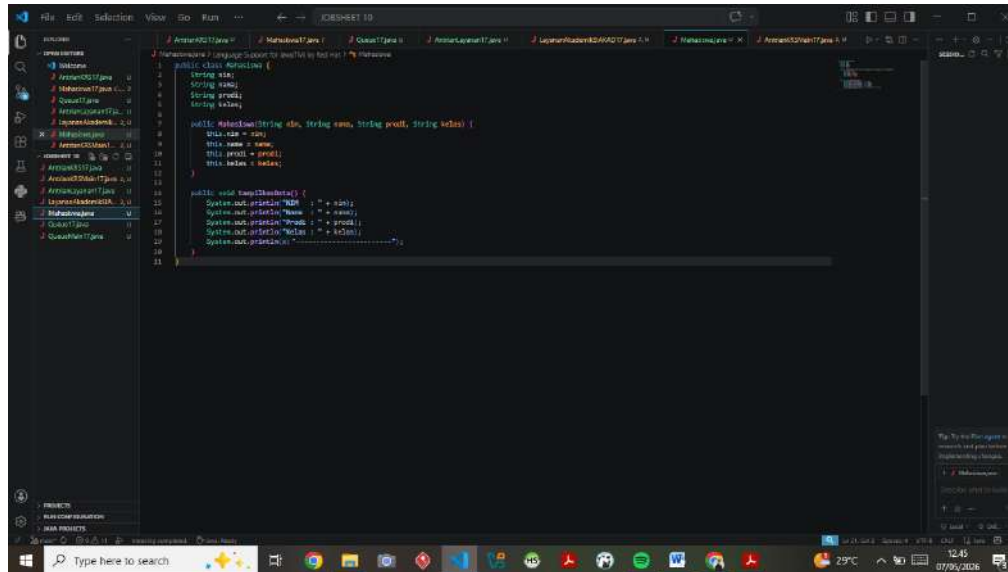
```
antrian.lihatAkhir();
break;
```



```
3 public class LayananAkademikSIKAD17 {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         while (true) {
8             menu();
9
10            int pilih = sc.nextInt();
11
12            switch (pilih) {
13                case 1:
14                    tambahMahasiswa();
15                    break;
16                case 2:
17                    layaniMahasiswa();
18                    break;
19                case 3:
20                    cekMahasiswaTerdepan();
21                    break;
22                case 4:
23                    tampilkanSemuaAntrian();
24                    break;
25                case 5:
26                    jumlahMahasiswa();
27                    break;
28                case 6:
29                    antrian.lihatAkhir();
30                    break;
31            }
32
33            if (pilih == 0) {
34                break;
35            }
36        }
37
38        sc.close();
39    }
40 }
```


TUGAS – ANTRIAN KRS

Class Mahasiswa



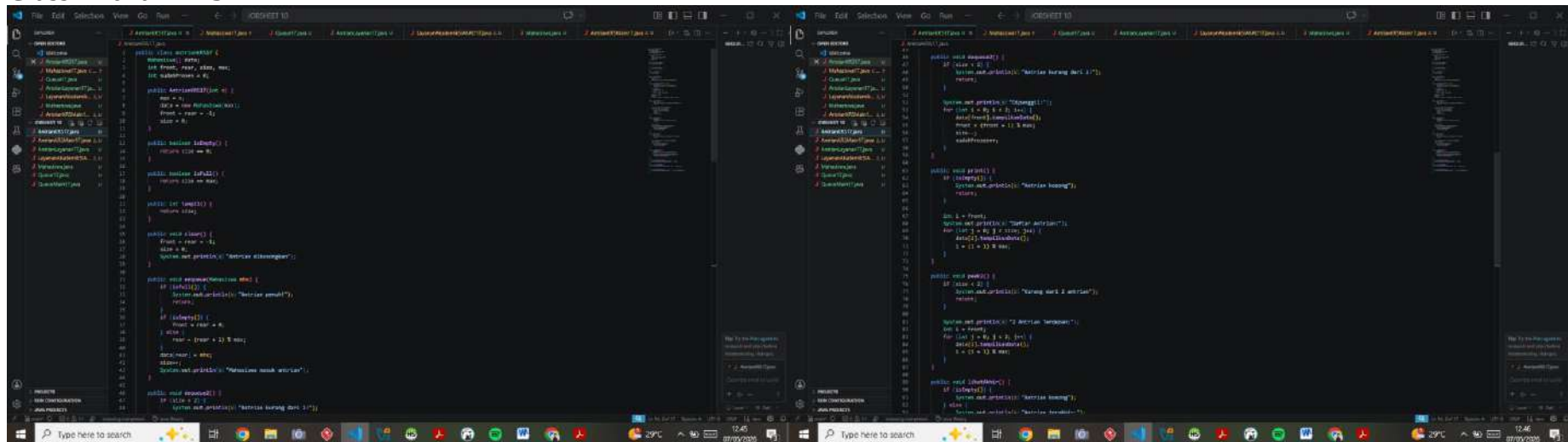
```
import java.util.Scanner;

public class Mahasiswa {
    String nama;
    String nim;
    String prodi;
    String kelas;

    public Mahasiswa(String nama, String nim, String prodi, String kelas) {
        this.nama = nama;
        this.nim = nim;
        this.prodi = prodi;
        this.kelas = kelas;
    }

    public void tampilData() {
        System.out.println("Nama : " + nama);
        System.out.println("Nim : " + nim);
        System.out.println("Prodi : " + prodi);
        System.out.println("Kelas : " + kelas);
        System.out.println("-----");
    }
}
```

Class AntrianKRS



```
import java.util.Scanner;

public class AntrianKRS {
    int front, rear, elem, max;
    int antrianKRS[];

    public AntrianKRS(int n) {
        max = n;
        antrianKRS = new int[max];
        front = rear = -1;
    }

    public boolean isEmpty() {
        return front == rear;
    }

    public boolean isFull() {
        return (front == rear);
    }

    public void enqueue() {
        if (isFull()) {
            System.out.println("Antrian sudah penuh");
        } else {
            rear = rear + 1;
            antrianKRS[rear] = elem;
            System.out.println("Antrian sudah masuk");
        }
    }

    public void dequeue() {
        if (isEmpty()) {
            System.out.println("Antrian sudah kosong");
        } else {
            front = front + 1;
            elem = antrianKRS[front];
            System.out.println("Antrian sudah keluar");
        }
    }

    public void tampilData() {
        if (isEmpty()) {
            System.out.println("Antrian sudah kosong");
        } else {
            System.out.println("Antrian sudah masuk");
        }
    }
}
```

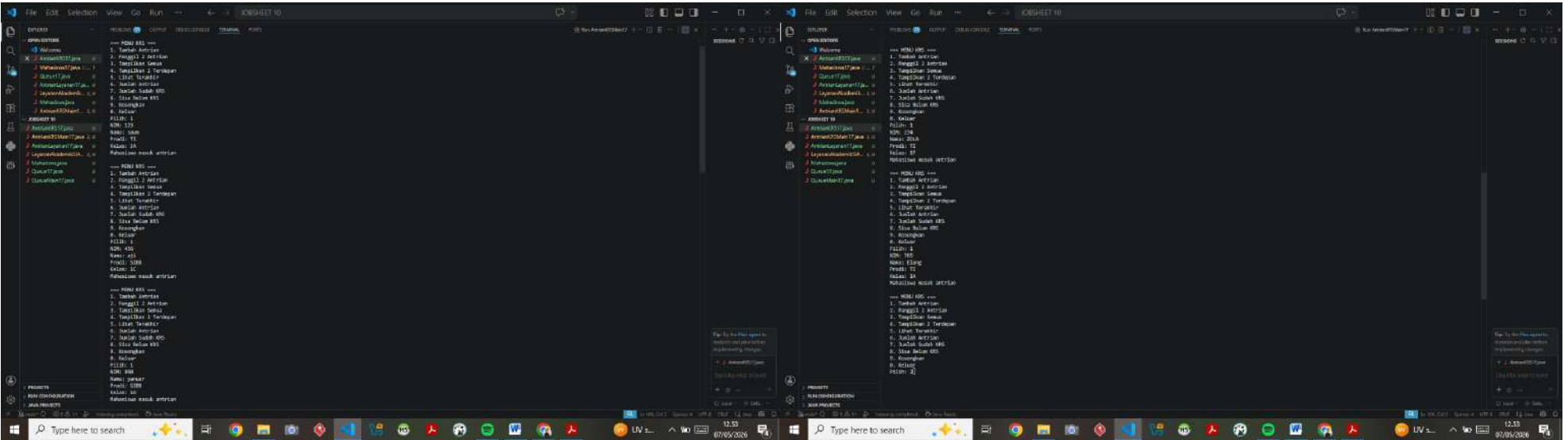
```
1 public void enqueue() {  
2     if (size < 2) {  
3         System.out.println("Masuk dari 2 antrian");  
4         return;  
5     }  
6     System.out.println("Antrian Terpenuhi");  
7     int k = front;  
8     for (int i = k; i < 2; i++) {  
9         data[i] = data[k];  
10        k = (i + 1) % 5;  
11    }  
12    public void libatkan() {  
13        if (isEmpty()) {  
14            System.out.println("Tidak kosong");  
15            return;  
16        }  
17        System.out.println("Membaca barisan");  
18        int k = front;  
19        while (k != rear) {  
20            System.out.println("Barisan antrian: " + data[k]);  
21            k = (k + 1) % 5;  
22        }  
23    }  
24    public void dequeue() {  
25        System.out.println("Barisan antrian: " + data[front]);  
26        front = (front + 1) % 5;  
27    }  
28    public void print() {  
29        System.out.println("Barisan antrian: " + data[front]);  
30    }  
31 }
```

Class AntrianKRSMain

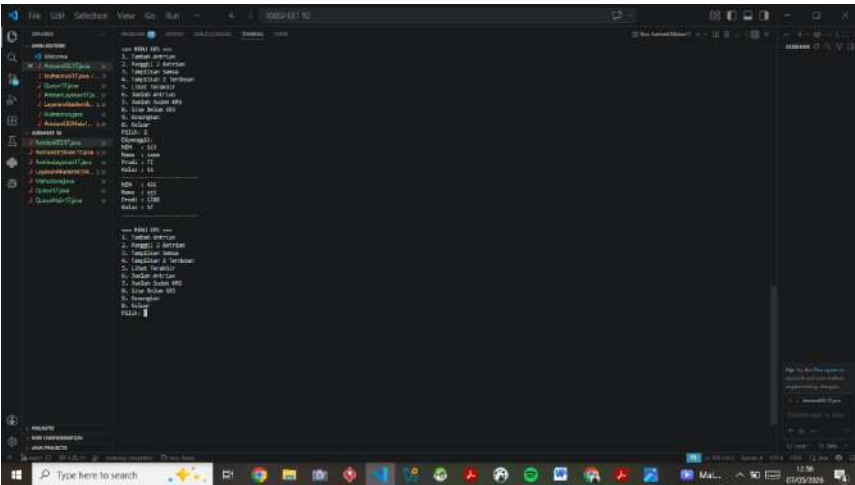
```
1 import java.util.Scanner;  
2  
3 public class AntrianKRSMain {  
4     Scanner sc = new Scanner(System.in);  
5     AntrianKRS antrian = new AntrianKRS(5);  
6  
7     public void enqueue() {  
8         System.out.println("Masuk dari 2 antrian");  
9         return;  
10    }  
11    public void dequeue() {  
12        System.out.println("Barisan antrian: " + data[front]);  
13        front = (front + 1) % 5;  
14    }  
15    public void print() {  
16        System.out.println("Barisan antrian: " + data[front]);  
17    }  
18    public void libatkan() {  
19        if (isEmpty()) {  
20            System.out.println("Tidak kosong");  
21            return;  
22        }  
23        System.out.println("Membaca barisan");  
24        int k = front;  
25        while (k != rear) {  
26            System.out.println("Barisan antrian: " + data[k]);  
27            k = (k + 1) % 5;  
28        }  
29    }  
30    public void enqueue() {  
31        System.out.println("Masuk dari 2 antrian");  
32        return;  
33    }  
34    public void dequeue() {  
35        System.out.println("Barisan antrian: " + data[front]);  
36        front = (front + 1) % 5;  
37    }  
38    public void print() {  
39        System.out.println("Barisan antrian: " + data[front]);  
40    }  
41    public void libatkan() {  
42        if (isEmpty()) {  
43            System.out.println("Tidak kosong");  
44            return;  
45        }  
46        System.out.println("Membaca barisan");  
47        int k = front;  
48        while (k != rear) {  
49            System.out.println("Barisan antrian: " + data[k]);  
50            k = (k + 1) % 5;  
51        }  
52    }  
53 }
```

```
1 public void enqueue() {  
2     System.out.println("Masuk dari 2 antrian");  
3     return;  
4 }  
5  
6 public void dequeue() {  
7     System.out.println("Barisan antrian: " + data[front]);  
8     front = (front + 1) % 5;  
9 }  
10  
11 public void print() {  
12     System.out.println("Barisan antrian: " + data[front]);  
13 }  
14  
15 public void libatkan() {  
16     if (isEmpty()) {  
17         System.out.println("Tidak kosong");  
18         return;  
19     }  
20     System.out.println("Membaca barisan");  
21     int k = front;  
22     while (k != rear) {  
23         System.out.println("Barisan antrian: " + data[k]);  
24         k = (k + 1) % 5;  
25     }  
26 }  
27  
28 public void enqueue() {  
29     System.out.println("Masuk dari 2 antrian");  
30     return;  
31 }  
32  
33 public void dequeue() {  
34     System.out.println("Barisan antrian: " + data[front]);  
35     front = (front + 1) % 5;  
36 }  
37  
38 public void print() {  
39     System.out.println("Barisan antrian: " + data[front]);  
40 }  
41  
42 public void libatkan() {  
43     if (isEmpty()) {  
44         System.out.println("Tidak kosong");  
45         return;  
46     }  
47     System.out.println("Membaca barisan");  
48     int k = front;  
49     while (k != rear) {  
50         System.out.println("Barisan antrian: " + data[k]);  
51         k = (k + 1) % 5;  
52     }  
53 }
```

Menu 1



Menu 2



[illegible][illegible]

The screenshot shows a Windows desktop with a dark theme. A Notepad++ window is open, displaying a list of 10 Indonesian islands. The list is organized into two sections: 'Jawa' and 'Jawa Barat'. The first section lists 10 islands, and the second section lists 10 islands. The desktop background is dark, and the taskbar at the bottom shows various application icons and the system clock.

Notepad++ Content:

```

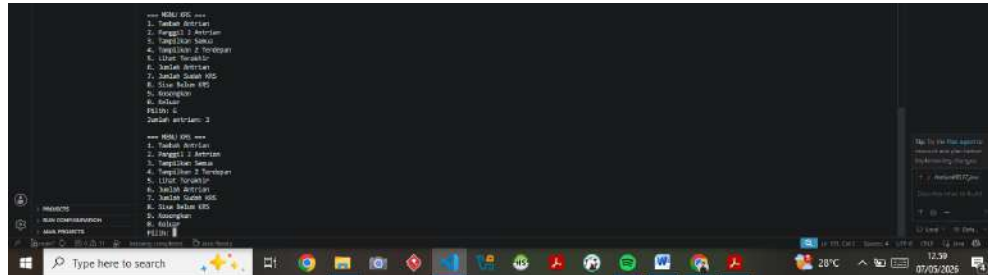
===== Jawa =====
1. Sumatra
2. Sumatra Barat
3. Sumatra Tengah
4. Sumatra Selatan
5. Sumatra Utara
6. Sumatra Barat
7. Sumatra Tengah
8. Sumatra Selatan
9. Sumatra Utara
10. Sumatra Barat

===== Jawa Barat =====
1. Sumatra
2. Sumatra Barat
3. Sumatra Tengah
4. Sumatra Selatan
5. Sumatra Utara
6. Sumatra Barat
7. Sumatra Tengah
8. Sumatra Selatan
9. Sumatra Utara
10. Sumatra Barat
  
```

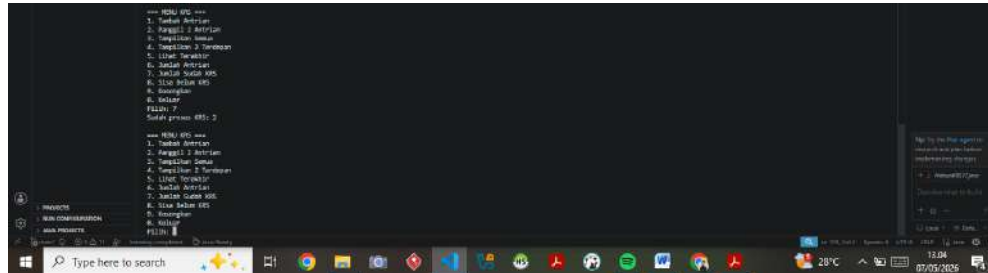
Taskbar and System Tray:

- Taskbar icons: Start button, Search bar, File Explorer, Edge, Chrome, VS Code, Notepad++, and several other application icons.
- System Tray: Date and Time (07/05/2025, 11:00), Network status, Volume, and other system indicators.

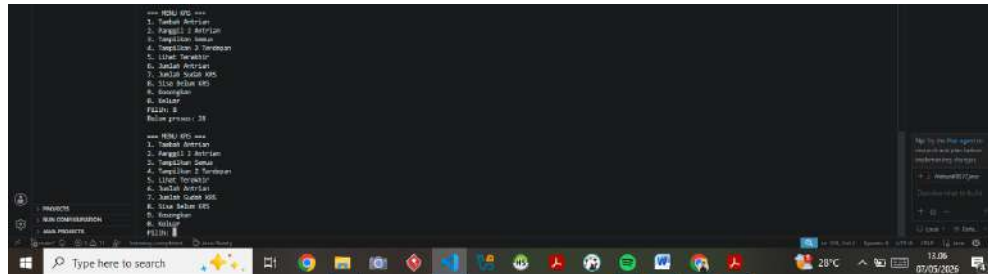
Menu 6



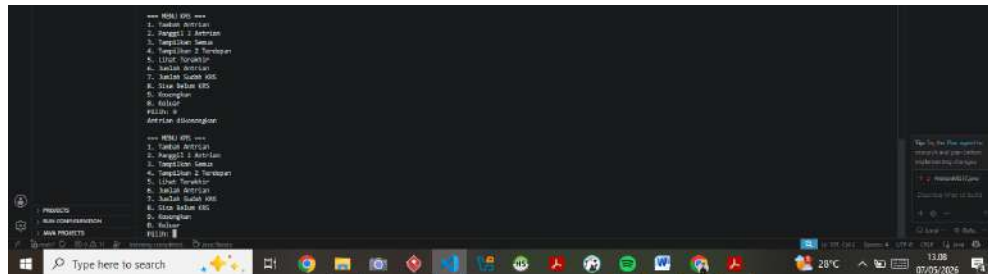
Menu 7



Menu 8



Menu 9



LinkGitHub: <https://github.com/saumsaifulloh/AlgoritmaDanStrukturDataSem2/tree/main/ASD/PASD/JOBSHEET%2010>